

Gaurav Dixit

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Machine Learning and robotics researcher specializing in multi-objective multiagent decision-making, adaptive coordination under uncertainty, and human-centered autonomy. Experienced in developing learning algorithms and simulation-based validation methods for coordinated behavior, dynamic trade-offs and generalization.

TECHNICAL SKILLS

AI / Robotics : Reinforcement Learning, Multi-Objective Optimization, Cooperative Decision-Making, Evolution Strategies,

Tools : Pytorch / LibTorch, Numpy, JAX, MuJoCo, Tensorflow, Matplotlib

Languages : C/C++, Python, MATLAB, Haskell, Rust, TypeScript

EDUCATION

Oregon State University

Sept. 2021 – Sept. 2023

Doctor of Philosophy (Ph.D.) in Robotics

Corvallis, OR

- *Learning to coordinate in sparse asymmetric multiagent systems*
- *Advised by Dr. Kagan Tumer*

Oregon State University

Sept. 2018 – Sept. 2021

Master of Science (M.S.) in Computer Science

Corvallis, OR

- *Dirichlet-multinomial counterfactual rewards for heterogeneous multiagent systems*
- *Advised by Dr. Kagan Tumer, Dr. Weng-Keen Wong*

Pune Institute of Computer Technology

May 2012 – May 2016

Honors Bachelor of Engineering (B.Eng.), magna cum laude

Pune, India

- *Stochastic Variational Inference for Robust Named Entity Recognition*

EXPERIENCE

NSF AI-CARING Institute

Nov. 2023 – Present

Research Associate

Corvallis, OR

- Developing multi-objective reinforcement learning methods for autonomous systems that must balance safety, task progress, human preferences, and long-term autonomy.
- Leading work within the NSF AI-CARING Institute on formalizing human-centered beneficence decision-making.
- Advising graduate students and working directly with funding agencies to report progress and outcomes.

Oregon State University

Sept. 2018 – Sept. 2023

Graduate Research Assistant

Corvallis, OR

- Developed reinforcement learning and evolutionary optimization algorithms for multi-agent coordination under sparse rewards, heterogeneous roles, and partially aligned objectives.
- Designed behavioral diversity and generalization methods to analyze policy failures, coordination breakdowns, and robustness across unseen scenarios.

RedLynx Oy

July 2019 – Oct. 2019

Research Scientist - Intern

Helsinki, Finland

- Investigated tree-based planning with reinforcement learning for sequential decision-making in normal form games.
- Developed quality-diversity methods for learning latent behavior representations to support automated policy exploration and diversity-driven search.

Ubisoft Entertainment SA
Artificial Intelligence Engineer

July 2016 – May 2018
Pune, IN

- Implemented policy-gradient methods to develop an end-to-end pipeline for automated game testing.
- Designed and developed a rigid-body physics engine to support online simulation and interactive testing tools.

BMC Software
Research Development Software Engineer - Intern

June 2015 – June 2016
Pune, IN

- Developed a real-time event analysis tool for root-cause diagnosis using the Rete algorithm for network automation.

TEACHING

Graduate Teaching

- Co-led a tutorial at ECAI 2024 on Multi-Objective Multiagent Decision-Making
- Co-organizer of the Multi-Objective Decision Making Workshop at IJCAI and ECAI.
- Contributed to teaching and course development at OSU: Multiagent Systems and Learning-Based Control.
- Advised three graduate students to publish at top AI conferences.

Undergraduate Teaching

- Designed and taught Machine Learning (2015), Data Structures and Algorithms (2015), Programming Paradigms with C++ (2014), Operating System Administration (2013), Functional Programming with Haskell (2013).

SELECTED PUBLICATIONS (full list on Google Scholar)

1. **Dixit, G.**, & Tumer, K. (2026). *Pareto-Guided Exploration for Multi-Objective Multiagent Learning*. To appear at AAMAS 2026.
2. Gonzalez, E., **Dixit, G.**, & Tumer, K. (2025). *Dynamic Influence for Coevolutionary Agents*. Proceedings of GECCO 2025, 368–376.
3. Iyer, S.*, Aydeniz, A. A., **Dixit, G.**, & Tumer, K. (2025). *Multiagent Quality-Diversity for Effective Adaptation*. In ECAI 2025 (pp. 3323–3330). IOS Press. <https://ebooks.iospress.nl/doi/10.3233/FAIA251201>
4. **Dixit, G.** & Tumer, K. (2024). *Objective-Informed Diversity for Multi-Objective Multiagent Coordination*. In ECAI 2024 (pp. 3660–3667). IOS Press. <https://ebooks.iospress.nl/doi/10.3233/FAIA240923>
5. **Dixit, G.** & Tumer, K. (2023). *Learning inter-agent synergies in asymmetric multiagent systems*. Proceedings of AAMAS 2023, 1569–1577. <https://dl.acm.org/doi/epdf/10.5555/3545946.3598812>
6. **Dixit, G.** & Tumer, K. (2022). *Balancing teams with quality-diversity for heterogeneous multiagent coordination*. Proceedings of GECCO Companion, 236–239.
7. **Dixit, G.** & Tumer, K. (2022). *Behavior Exploration and Team Balancing for Heterogeneous Multiagent Coordination*. Proceedings of AAMAS 2022. <https://par.nsf.gov/servlets/purl/10359619>
8. **Dixit, G.**, Gonzalez, E., & Tumer, K. (2022). *Diversifying behaviors for learning in asymmetric multiagent systems*. Proceedings of GECCO, 350–358.
9. **Dixit, G.**, Koll, C.*, & Tumer, K. (2021). *Heterogeneous agent coordination via adaptive quality diversity and specialization*. Proceedings of GECCO Companion, 95–96.
10. Olson, M. L., Nguyen, T.-V., **Dixit, G.**, et al. (2021). *Contrastive identification of covariate shift in image data*. 2021 IEEE Visualization Conference (VIS), 36–40. IEEE.
11. **Dixit, G.**, Airiau, S., & Tumer, K. (2020). *Gaussian processes as multiagent reward models*. AAMAS Conference Proceedings. <https://www.ifaamas.org/Proceedings/aamas2020/pdfs/p330.pdf>
12. **Dixit, G.**, Zerbel, N., & Tumer, K. (2019). *Dirichlet-multinomial counterfactual rewards for heterogeneous multiagent systems*. International Symposium on Multi-Robot and Multi-Agent Systems (MRS), 209–215.